

A Symmetric Solution for n-tuples in Indeterminate Diophantine Equations

Ernesto H. Prieto Peña

Escuela Superior de Economía (ESE)

Instituto Politécnico Nacional (IPN)

Class: 1973-1978

ernestohprieto@yahoo.com.mx

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Abstract

This paper presents a rigorous recursive method for finding integer solutions to the Pythagorean equation for n variables. By leveraging square factors p_n , the method allows for a systematic expansion that ensures all terms belong to the set of integers $\mathbb{Z}$. The symmetry of the formulation allows for an elegant extension to any number of dimensions.

1 Introduction

Based on the foundational work of Diophantus and the Pythagorean school, this document proposes a solution that ensures all terms belong to the set of integers $\mathbb{Z}$. The algorithm allows for the construction of n -tuples of squares whose sum is itself a perfect square.

2 The General Formula

For any integer $x > 2$, let p_n be a square factor of x^2 . The elements of the Pythagorean triplet are defined as follows:

- **First Cathetus:** x
- **Second Cathetus:** $Y = \frac{x^2 - p_n}{2\sqrt{p_n}}$
- **Hypotenuse:** $Z = \frac{x^2 + p_n}{2\sqrt{p_n}}$

Critical Condition for Parity

To ensure that Y and Z result in integers ($Y, Z \in \mathbb{Z}$):

1. p_n must be a square factor of x^2 such that $x^2 > p_n$.
2. If x^2 is even, the square factor p_n must necessarily be even as well.

3 Recursive Method for n Variables

The power of this method lies in its recursive nature. To extend the equation to n variables, we follow this rule: The hypotenuse Z_k obtained in step k becomes the first cathetus x_{k+1} for step $k + 1$.

3.1 Example: From 2 to 3 variables

Step 1: Let $x_1 = 12$ ($x_1^2 = 144$). Choosing $p_1 = 4$:

$$Y_1 = \frac{144 - 4}{4} = 35; \quad Z_1 = \frac{144 + 4}{4} = 37 \Rightarrow 12^2 + 35^2 = 37^2$$

Step 2 (Recursion): Now, let $x_2 = Z_1 = 37$ ($x_2^2 = 1369$). Choosing $p_2 = 1$:

$$Y_2 = \frac{1369 - 1}{2} = 684; \quad Z_2 = \frac{1369 + 1}{2} = 685$$

Final Result: $12^2 + 35^2 + 684^2 = 685^2$. This process can be repeated infinitely to add x_n terms.

How to Cite

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This research was first conceived and published online in **October 2006**. The mathematical core presented here preserves the original intellectual contribution, providing a reliable and verifiable method for the recursive generation of Pythagorean n -tuples.

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